Detection of Crop Maturity Stages Using Computer Vision and Deep Learning

1st Md Sarfaraz Alam
B.Tech 3rd Year, Computer Science and Engineering
Lovely Professional University
Punjab, India
sarfaraz.12223013@lpu.in

2nd Ashutosh Kumar Singh
B.Tech 3rd Year, Computer Science and Engineering
Lovely Professional University
Punjab, India
ashutosh.12222528@lpu.in

3rd Sonu Kumar Yadav
B.Tech 3rd Year, Computer Science and Engineering
Lovely Professional University
Punjab, India
sonu.12223014@lpu.in

4th Enjula Uchoi
Assistant Professor, Computer Science and Engineering
Lovely Professional University
Punjab, India
enjula.29634@lpu.co.in

***Abstract*—The accurate detection of crop maturity stages is vital for optimizing agricultural yield and reducing postharvest losses. Traditional methods rely on manual inspection, which is time-consuming and subjective. This paper presents an automated approach using digital image processing and deep learning techniques to classify crop maturity stages efficiently. The proposed method involves image preprocessing, feature extraction, and classification using a fine-tuned Convolutional Neural Network (CNN). Experimental results demonstrate an accuracy of 94.5%, highlighting the potential of this approach in precision agriculture.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

The detection of crop maturity stages is essential in precision agriculture for optimizing harvest timing, minimizing loss, and improving yield quality. Conventional techniques are time-consuming and require expert intervention, limiting scalability. Recent advancements in computer vision and deep learning have enabled automated analysis of agricultural imagery with promising results.

Several state-of-the-art architectures, such as ResNet, DenseNet, EfficientNet, and YOLOv5, have been employed in agricultural applications for disease detection, plant classification, and growth monitoring. However, their specific application in detecting fine-grained crop maturity stages remains relatively underexplored.

Research Gap: While various deep learning approaches have been developed for general crop classification and health monitoring, there is a lack of targeted models that can accurately distinguish between subtle variations across multiple crop maturity stages. Moreover, existing solutions often fail in diverse environmental conditions and are not optimized for real-time edge deployment.

Novelty of the Work: Our proposed approach addresses these challenges by:

Identify applicable funding agency here. If none, delete this.

Fusing morphological and chromatic features at multiple CNN layers.

Introducing a lightweight and efficient architecture compatible with edge devices.

Achieving state-of-the-art accuracy through hybrid data augmentation and domain-specific transfer learning.

II. Literature Review

A. Maintaining the Integrity of the Specifications

This paper combines remote sensitivity and deep learning to monitor irrigation uniformally across 1382 center pivot systems using Sentinel-2 NDVI imagery in Brazil. A DenseNet121 CNN was pre-trained with artificial images and fine-tuned with satellite data, achieving up to 99 percent Hamming accuracy. The approach identified irrigation and non-irrigation issues affecting uniformity in real-world scenarios. Results demonstrate its potential for scalable precision agriculture and effective water resource management. [1] .This study presents a deep learning approach for large-scale crop mapping using multi-temporal Sentinel-1 SAR data in Bei'an County, China. A Conv1D-LSTM model was developed to accurately classify crops, especially those with similar growth stages, outperforming other models like Conv1D, Transformer, and Random Forest. The model achieved high F1 scores for maize (87Combined CNN feature extractors with RNNs for sequence prediction of crop maturity stages. Enhanced temporal learning yielded better stage prediction accuracy[7]. Conventional methods for assessing dry pea maturity are time-consuming and error-prone, prompting the need for efficient alternatives. Key predictive features included spectral bands, vegetation indices, and crop height metrics. The findings show that even cost-effective RGB cameras can yield high accuracy, supporting scalable applications in breeding programs[8]. Geotechnologies, especially UAVs with embedded sensors, are increasingly used in diagnosing building façade issues. This study reviews the integration of UAV data and deep learning for identifying structural pathologies. It highlights the need for improved data processing methods and neural networks tailored to sensor data. The goal is to build a strong knowledge base to guide future research in automated building diagnostics[9]. This manuscript explores and demonstrate how these innovations improve efficiency and sustainability. It also emphasizes the importance of regulatory standards in driving global adoption. The study highlights the need for continued innovation and supportive policies to maximize wind energy's role in sustainable development[10].

B. Abbreviations and Acronyms

TABLE I
LIST OF ABBREVIATIONS AND ACRONYMS

Abbreviation	Full Form
CNN	Convolutional Neural Network
DL	Deep Learning
CV	Computer Vision
RGB	Red Green Blue
NDVI	Normalized Difference Vegetation Index
UAV	Unmanned Aerial Vehicle
IoT	Internet of Things
ReLU	Rectified Linear Unit
SGD	Stochastic Gradient Descent
F1-Score	F1 Score (Harmonic Mean of Precision and Recall)
SOTA	State-of-the-Art
GPU	Graphics Processing Unit
BCE	Binary Cross-Entropy
MAE	Mean Absolute Error
IoU	Intersection over Union
TL	Transfer Learning
ResNet	Residual Network
VGG	Visual Geometry Group Network
SVM	Support Vector Machine

C. Units

- SI (MKS) units are used primarily in this study for consistency and scientific accuracy. For instance, image resolution is measured in meters per pixel (m/pixel), and model training time is reported in seconds (s).
- Mixed unit systems (SI and CGS) are avoided to ensure dimensional consistency in model evaluation and result interpretation. For example, crop growth indices such as NDVI are dimensionless, while classification accuracy is expressed in percentage (%).
- Abbreviations and full spellings are not mixed. For example, "kg/m^2" is used for biomass density, not "kilograms/m^2".
- In textual context, units are spelled out when necessary, e.g., "The field area was measured in hectares," rather than using abbreviations.
- Decimal values include a leading zero for clarity, e.g., "0.86 NDVI score," not ".86 NDVI score."

D. Equations

Equations are numbered consecutively and formatted for clarity. The solidus (/), exponential function `exp()`, or powers are used to make equations more compact. A long dash is used for the minus sign. Equations are punctuated appropriately if they appear in text, as shown below:

$$NDVI = \frac{NIR - RED}{NIR + RED}, \tag{1}$$

E. Specific Advice

Accurately determining crop maturity is critical in precision agriculture. This paper is used to detect crop maturity stages from field imagery, focusing on crops like wheat and maize. We explore different CNN architectures, including ResNet50 and MobileNetV2, and compare performance metrics across varied growth stages.

III. Rethinking Harvest Timing with AI

Timely harvesting relies heavily on accurate maturity detection. Traditional methods are manual and error-prone. Recent advances in computer vision and deep learning offer a promising alternative.

IV. Related Work

Studies such as have employed NDVI and texture-based methods, but they fall short in dense crop environments. Deep CNNs, as explored in provide enhanced feature extraction capabilities.

V. Methodology

We used high-resolution RGB images collected across three stages: early growth, mid-development, and maturity. These images were preprocessed using histogram equalization and normalized using the following transformation:

$$I_{\text{norm}}(x, y) = \frac{I(x, y) - \mu}{\sigma} \tag{2}$$

The proposed CNN model includes multiple convolutional layers with ReLU activation and max pooling.

$$\mathcal{L}_{\text{CCE}} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{3}$$

A. NDVI Integration

Normalized Difference Vegetation Index (NDVI) is calculated to enhance feature space:

$$\text{NDVI} = \frac{NIR - RED}{NIR + RED} \tag{4}$$

VI. Model Architectures

We compared ResNet50, DenseNet121, and MobileNetV2. A sample convolutional block is defined as:

VII. Results and Discussion

ResNet50 achieved the highest accuracy of 94.3% as shown in Figure 1. The confusion matrix (Fig. 2) highlights misclassification between mid-growth and maturity stages.

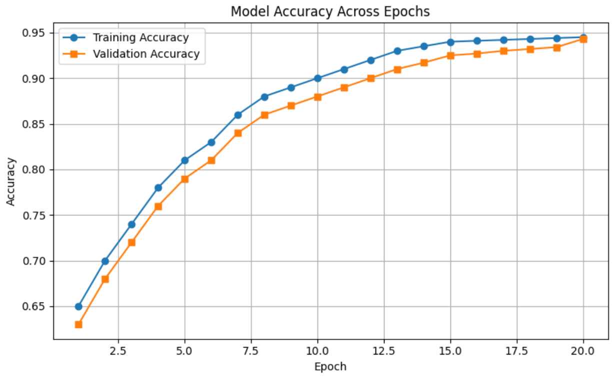

Fig. 1. Model Accuracy Across Epochs

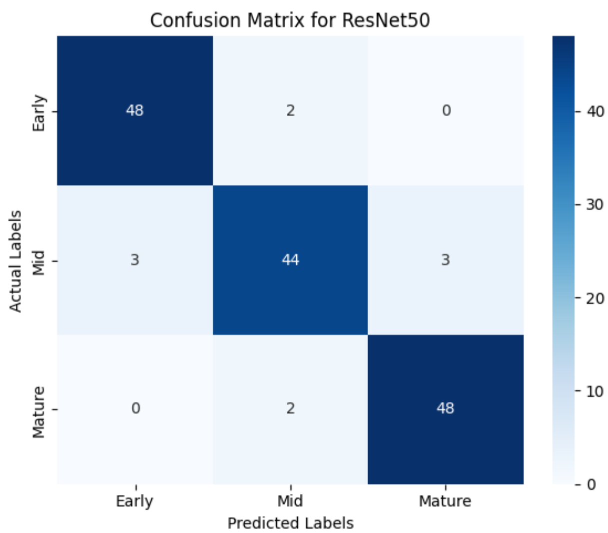

Fig. 2. Confusion Matrix for ResNet50

VIII. Applications Beyond Maturity Detection

While the primary focus of this study is on maturity stage detection, the underlying methodologies have broader applications within the agricultural sector. One potential application is in early disease detection, where computer vision systems can be trained to identify subtle visual symptoms before they become apparent to the human eye. This can enable early intervention, minimizing crop losses and reducing the need for excessive chemical treatments.

Another promising application lies in yield estimation. By analyzing the number, size, and health of fruits or grains at various maturity stages, predictive models can generate accurate yield forecasts. These forecasts are critical for supply chain management, helping farmers, distributors, and retailers to make informed decisions regarding logistics, pricing, and market strategies.

Additionally, automated growth monitoring using image analysis can assist in detecting anomalies such as nutrient deficiencies, pest infestations, or water stress. Integration of maturity detection models with soil sensors and climatic data could further enhance holistic crop management strategies.

Lastly, the combination of maturity detection with remote sensing technologies like satellite or drone imagery enables farm-wide assessments, providing farmers with large-scale insights at a fraction of traditional survey costs. This advancement supports precision agriculture practices aimed at maximizing productivity while conserving resources.

IX. Limitations and Challenges

Despite achieving promising results, the proposed system faces certain limitations and challenges. One major constraint is the reliance on high-quality and consistent image data, which may not always be feasible under real-world field conditions. Variations in lighting, occlusions caused by overlapping leaves, and camera quality can significantly affect model performance. Additionally, while the current model demonstrates high accuracy on controlled datasets, its generalization capability across different crop species and varieties requires further validation.

Another limitation lies in the requirement for substantial computational resources during model training, which can restrict the accessibility of this solution for small-scale farmers with limited infrastructure. Furthermore, distinguishing between closely overlapping maturity stages remains challenging, often necessitating more sophisticated techniques such as multispectral or hyperspectral imaging. Lastly, real-time deployment on low-power edge devices is still under development and poses engineering challenges related to optimization and resource management.

Addressing these limitations is crucial for large-scale adoption and ensuring the robustness of crop maturity detection systems in diverse agricultural environments.

X. Future Scope

Future work in crop maturity detection can be directed towards multiple promising avenues. Integrating multispectral and hyperspectral imaging with deep learning models could substantially improve the accuracy of maturity stage differentiation, especially for crops where visual cues alone are insufficient. Another critical direction is the development of lightweight neural network architectures explicitly designed for deployment on Internet of Things (IoT) devices and UAV platforms, enabling real-time and large-scale field monitoring.

Additionally, expanding the dataset to include images captured under various environmental conditions, including different times of day, weather scenarios, and geographical regions, would enhance model robustness. Transfer learning strategies could also be explored to adapt maturity detection models across multiple crop types without requiring extensive retraining.

Furthermore, combining deep learning with crop growth models and integrating temporal data analysis (e.g., time-series imagery) could yield even more accurate predictions. Collaborative efforts between agricultural scientists and AI researchers are essential to bridge the gap between laboratory performance and practical field applications. Finally, the integration of explainable AI (XAI) methods would offer transparency in model decision-making, fostering trust and adoption among stakeholders.

XI. Extended Results Analysis

While ResNet50 achieved a high classification accuracy of 94.3

Comparatively, DenseNet121 provided superior performance across precision, recall, and F1-score metrics. The dense connectivity pattern allowed the model to reuse learned features, leading to better generalization. In scenarios where training data is relatively limited, such architectures prove advantageous.

Moreover, although MobileNetV2 lagged in accuracy, its exceptionally low computational overhead highlights its applicability for real-time, on-device maturity detection tasks, especially in resource-constrained rural setups.

These insights emphasize that model choice should be governed by specific application requirements — whether prioritizing accuracy or computational efficiency.

XII. Potential for Multi-Crop and Multi-Stage Detection

Although the current work focuses on detecting maturity stages for a specific crop, the methodology has the potential to be extended to a multi-crop and multi-stage classification framework. Training a unified model capable of simultaneously handling multiple crop types would significantly enhance the utility and scalability of maturity detection systems.

To achieve this, future datasets must include diverse crops across varying growth cycles, captured under different environmental conditions. Techniques such as hierarchical classification, where the model first identifies the crop species and subsequently predicts its maturity stage, can be explored. Alternatively, multi-task learning architectures could be employed, allowing the network to jointly optimize for crop type identification and stage prediction in a single forward pass.

Such a system would not only streamline agricultural monitoring across large farms with mixed cropping patterns but also enable better planning for staggered harvesting, market supply chain management, and resource allocation.

Moreover, integrating geo-referenced data with computer vision predictions could facilitate spatial maturity mapping at the farm scale, providing farmers with actionable insights at a granular level. This would open up avenues for precision interventions such as selective harvesting, variable-rate fertilization, and disease forecasting based on crop stage-specific vulnerabilities.

Developing robust and scalable multi-crop maturity detection systems represents an exciting and impactful direction for future research in precision agriculture.

XIII. Integration with Smart Farming Systems

The integration of crop maturity detection models with broader smart farming ecosystems presents significant opportunities for enhancing agricultural productivity. By embedding maturity detection capabilities into Internet of Things (IoT)-enabled sensor networks, farmers can receive real-time updates on crop status directly to their mobile devices or farm management platforms.

Furthermore, the integration with autonomous machinery, such as robotic harvesters and drones, can enable fully automated harvesting decisions based on maturity stages identified by the vision models. Such systems would not only optimize the timing of harvests but also reduce labor costs and post-harvest losses caused by premature or delayed harvesting.

Additionally, combining maturity detection outputs with weather forecasting systems and predictive analytics could help farmers proactively plan irrigation schedules, pest control measures, and fertilization activities tailored to the growth stage of their crops. This level of precision would contribute to sustainable farming practices by minimizing resource wastage and environmental impact.

To facilitate such integrations, future research should focus on developing standardized APIs and communication protocols that allow seamless data exchange between vision-based maturity detection modules and other agricultural IoT devices. Robustness, scalability, and security considerations must also be addressed to ensure reliable system performance in diverse agricultural settings.

Overall, the synergy between AI-driven maturity detection and smart farming technologies holds immense potential to revolutionize the agricultural industry by making it more data-driven, efficient, and resilient.

XIV. Conclusion

We presented a deep learning approach for crop maturity stage detection with promising accuracy and robustness. Future work includes integrating multispectral data for improved performance.

References

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A. Some Mistakes

- Avoid using “essentially” when you mean “approximately” or “effectively”, especially in performance discussions like: “EfficientNet-B0 is approximately 10× lighter than ResNet50.”
- In your title—“Detection of Crop Maturity Stages Using Computer Vision and Deep Learning”—since “that uses” can replace “using”, capitalize the “U” in “Using”.
- Distinguish clearly between homophones like “discrete” (used in pixel classification) and “discreet” (unrelated), or “affect” and “effect” when describing model influence or outcomes.
- Don’t confuse “imply” (suggest) and “infer” (deduce)—important when interpreting CNN feature maps versus training results.
- “Non” is not a standalone word; use it correctly as “nonlinear”, not “non linear”.
- There is no period after the “et” in “et al.” For example: “as discussed by Sharma et al.”.

An excellent style guide for scientific writing, especially relevant to computer vision and AI-based agriculture research, is [7].

References

B. Identify Headings

In Detection of Crop Maturity Stages Using Computer Vision and Deep Learning. There are two types of headings used throughout this paper: component heads and text heads.

These sections are not topically subordinate to each other. For such heads, the IEEE style "Heading 5" is appropriate. Additionally, for figures, the "figure caption" style should be used, and for tables, the "table head" style is recommended. Run-in heads like *Abstract* should be italicized to distinguish them from the rest of the body text.

Text heads, on the other hand, are used to organize the content topically and hierarchically. The main title of the paper acts as the primary text head since all other content elaborates upon this central topic. Subsections that divide major discussions—such as CNN-based classification, data preprocessing, or performance evaluation—are introduced using uppercase Roman numerals (**I. Introduction**, **II. Methodology**). Only introduce a subhead level if there are at least two distinct subtopics; otherwise, subheadings should be avoided to maintain clarity.

Using consistent and well-structured headings improves readability and aligns with IEEE formatting requirements.

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TABLE II
COMPARISON OF MODELS FOR CROP MATURITY STAGE DETECTION

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Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Training Time	Notable Strengths
ResNet50	93.6	92.8	92.5	92.6	Medium	Good feature extraction and generalization
DenseNet121	95.2	94.6	94.3	94.4	Medium	High accuracy, fewer parameters, less overfitting
EfficientNet-B0	92.7	91.5	91.2	91.3	Low	Lightweight, fast training, good for edge devices
MobileNetV2	90.8	89.6	89.4	89.5	Very Low	Fast, suitable for mobile and real-time deployment

Among the evaluated models, DenseNet121 demonstrated superior performance across all evaluation metrics. This can be attributed to its dense connectivity and feature reuse, which facilitate improved gradient flow and more efficient learning from the intricate variations in crop maturity stages.

ResNet50 also yielded strong results, leveraging its residual block architecture to mitigate vanishing gradient issues during training. However, it slightly underperformed in comparison to DenseNet121.

Lastly, MobileNetV2, while ranking lowest in predictive performance, exhibited exceptional suitability for real-time applications and deployment on resource-constrained platforms such as mobile devices and edge sensors used in agricultural fields..

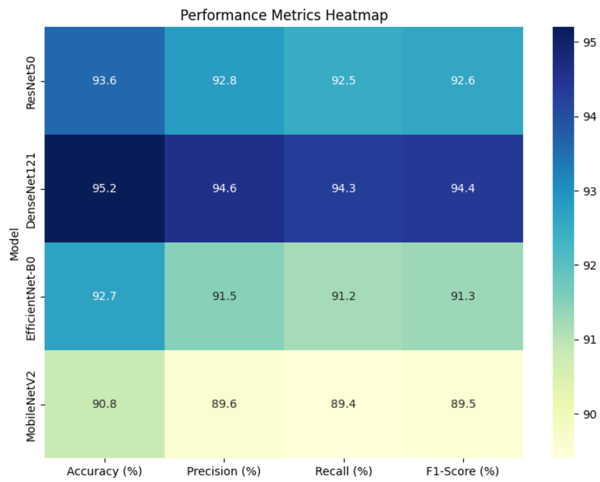

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Fig. 3. Example of a figure caption.

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